Minsuh Joo

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Seoul, South Korea

Education

- **Ewha Womans University** *Mar. 2023 - Jul. 2027*
  B.S. in Artificial Intelligence Seoul, South Korea
  - GPA: 4.28/4.3

Publications

P=Preprint, W=Workshop, B=Book

[P.1] **Minsuh Joo**, Youna Kim, Choonghyun Park, Soo Kyung Kim, Hyunsoo Cho. (2026). **MAPLE: A Multi-Axis Parametric-knowledge extraction Lens for LLM Evaluation**. *arXiv preprint*. **[Paper] [Code]**

[W.1] **Minsuh Joo**, Hyunsoo Cho. (2025). **Cleanse: Uncertainty Estimation Approach Using Clustering-based Semantic Consistency in LLMs**. *In Proceedings of the GEM Workshop at ACL 2025*. **[Paper] [Code]**

[B.1] Yoonseo Chung, Jeonghyun Kim, MiYeon Kim, **Minsuh Joo**, Hyunsoo Cho. (2025). **Foundations of LLMs and Financial Applications**. *Finance and Large Language Models, Springer Nature*, ISBN: 978-981-96-5832-9.

Research Experience

- **LKI Lab** *Jun. 2024 - Present*
  Undergraduate Research Assistant Ewha Womans University
  - Proposed a Dunn's Index-inspired semantic clustering metric for LLM uncertainty estimation, outperforming perplexity, LN-Entropy, and lexical similarity baselines in AUROC.
  - Developed MAPLE, a 1.3K-example benchmark and multi-axis evaluation framework for assessing knowledge extraction enhancement across Llama, Qwen, and OLMo models.
  - Investigated answerability-aware abstention and developed an SFT- and RL-based approach to promote evidence-grounded responses and selective refusal.

Selected Projects

- **Option-Specific RAG for LLM Question Answering** *Oct. 2025 - Dec. 2025*
  Python, FAISS, PyTorch **[Code]**
  - Designed an option-specific RAG pipeline with question-type-aware prompting and majority-voting ensembles to improve evidence retrieval and answer reliability.
  - Achieved 88% validation accuracy and 81% test accuracy, securing 3rd place in the competition.

- **PuBaO: Psychological Counseling Voiceover System** *Mar. 2026 - Aug. 2026*
  Python, Gemini Flash, GPT-SoVITS, Silero VAD, whisper, AudioWorklet **[Code]**
  - Designed a low-latency voice interaction pipeline integrating VAD, STT, LLMs, and TTS for real-time counseling.
  - Developed a dual-role LLM workflow separating response generation from dialogue assessment to enable immediate responses and subsequent question generation.

- **CORDE: Outfit Recommendation Chatbot** *Sep. 2025 - Feb. 2026*
  Python, FastAPI, FAISS **[Code]**
  - Constructed separate style- and occasion-based retrieval databases using fashion product images and metadata.
  - Designed a two-stage recommendation pipeline combining FAISS-based retrieval, weighted scoring, and LLM-based reranking to generate outfit recommendations.

Honors and Awards

- **LG Aimers 8th AI Hackathon: 3rd Place** *2026*
  LG AI Research, DACON
- **National Science and Engineering Scholarship** *2025-2026*
  Korea Student Aid Foundation (KOSAF)
- **Academic Excellence Awards – Department of Artificial Intelligence** *2025–2026*
  Ranked 1st in the major for three semesters, Ewha Womans University
- **MIXUP AI Datathon 3rd Prize: Upstage Solar API Prompt Engineering** *2025*
  Prometheus, Tobigs, BITAmin
- **Academic Excellence Awards – College Level** *2024*
  Ranked 1st and 2nd in the college in separate semesters, Ewha Womans University
- **Undergraduate Research Opportunities Program (UROP) Scholarship** *2024*
  Ewha Womans University

Skills

- **Programming:** Python, C, SQL
- **ML Frameworks:** PyTorch, HuggingFace
- **Libraries & Tools:** NumPy, pandas, FAISS, LaTeX

Experience

- **Prometheus** *Sep. 2025 - Feb. 2027*
  Team Member Union AI Club
  ◦ Led the Development Team as department head, managing several projects and coordinating the organizational operations.
- **Department of Artificial Intelligence Student Council** *Mar. 2023 - Dec. 2024*
  Planning Committee Member Ewha Womans University
  ◦ Organized academic and social events for AI majors, including orientation and school festival.